# CCCA Cell-State Analysis — Li et al. 2019, Skin

*Cutaneous Melanoma · MARS-seq · 1,082 malignant cells (3 samples ≥ threshold)*

## GO / NO-GO GATES

### Gate 1 — mixing & continuity GO

Harmony θ-sweep read on the regenerated embedding + per-neighbourhood iLISI (never a scalar): the kNN graph is ONE connected component at every θ∈{2,5,10,20} × n_pcs∈{20,30,40}; iLISI rises 1.30→1.80/3 and plateaus (θ=10 settled). Each of the 3 patients spans BOTH density lobes (p21 43/57, p25 59/41, p26 58/42) — a patient-shared axis, not an archipelago. The two visual poles are biologically identical (signatures + cell cycle equal), so they are density poles of one manifold.

### Gate 2 — module geography GO (non-branching)

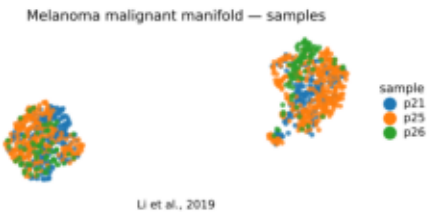
Low-signal melanoma (raw means Gast −0.015, pSom +0.016, Crest −0.002). Gast & pSom co-localise at the origin and decline along pseudotime (Gast ρ=−0.47 p=2e-60; pSom flat ρ=+0.05) opposed to a terminal neural-crest end that rises (Crest ρ=+0.59 p=6e-101; Crest-high decile pt=1.00). Crest is anti-correlated with both modules in space (Crest↔Gast −0.37, Crest↔pSom −0.43); joint Gast&pSom-high enrichment 0.20 (<1) — an inverse terminal gradient, not a co-expression hub. Crest peaks at the terminus, not on a Gast arm — no inversion flag.

### Gate 3 — hub cross-repression NA

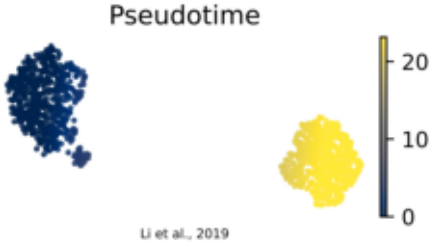
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks, fold-free self-approach 2.99; a PPT tree on the same manifold also yields 0 forks) and no Gast/pSom co-expression hub, so there is no branch pair to test for Gast↔pSom cross-repression.

## STEP FIGURES

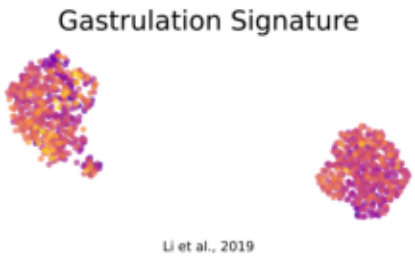
01 · Samples (Gate 1)




02 · Pseudotime




03 · Gastrulation



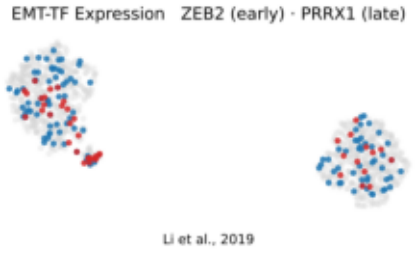

03 · Somitogenesis




03 · Neural Crest




04 · EMT-TFs




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**Dataset.** Melanoma malignant cells (Li et al., 2019), MARS-seq — an immune-focused atlas: 1,160 of 78,336 cells are malignant; ≥56-cell dominant-gap floor keeps 3/20 samples → 1,082 cells on the trajectory.

**Verdict.** Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous, sample-integrated gastrulation → terminal-neural-crest axis rooted at the Gast/pSom-high origin. G1 GO / G2 GO / G3 NA; no Puram-style hub-and-spoke bifurcation (curve 0 forks), so Gate 3 has no hub to test. EMT-TFs ZEB2 (body) / PRRX1 (sparse), detection-gated on this shallow dataset.

## PARAMETER CHOICES

| | |
|---|---|
| **Sample threshold** | 56 cells (dominant multiplicative gap 154→21, ratio 7.3×); 3/20 samples, 1,082 cells |
| **Overdispersed genes** | 217 (scFates find_overdispersed); no sc.pp.scale before PCA |
| **Integration** | Harmony on sample, θ=10 (iLISI plateau; strongly patient-private melanoma programs) |
| **Embedding** | n_pcs=30 (mid robust 20–40), n_neighbors=15; single elongated axis (aspect 7.5) |
| **Geometry** | Non-branching → scFates principal curve, Nodes=20, λ=0.3, μ=100 |
| **Topology check** | curve 2 tips / 0 forks, fold self-approach 2.99 (>1 fold-free); PPT tree 0 forks |
| **Root** | tip 0 = Gast/pSom-high, Crest-low origin (from developmental signature order) |
| **Module means (raw)** | Gast −0.015 · pSom +0.016 · Crest −0.002 (low-signal melanoma) |
| **Hub / r** | NA — no bifurcation hub (non-branching), no Gast↔pSom cross-repression r to report |
| **EMT-TFs shown** | ZEB2 (body) ↔ PRRX1 (sparse terminus); detection-gated; VIM excluded (not a TF) |